LaST-HD: Learning Latent Physical Reasoning from Human Data for Robot Manipulation

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Abstract: Human-hand demonstrations provide a direct and scalable source of physical interaction data for robot learning. While manual retargeting is indispensable for establishing kinematic action correspondence across different morphologies, robust transfer requires going beyond geometry to address the underlying alignment of physical dynamics between human and robot manipulation. To address this, we introduce LaST-HD, a novel human-to-robot action learning paradigm that extends reasoning-before-acting VLA by aligning human-hand and robot demonstrations in a shared latent reasoning space. Rather than mimicking human kinematics, LaST-HD trains an auxiliary action-conditioned world model on human-hand and robot trajectories to synthesize unified latent targets. After aligning cross-embodiment representations through predicted physical consequences, these targets supervise LaST-HD's latent reasoning process, enabling it to internalize shared physical dynamics and drive efficient human-hand action learning. Moreover, we develop Out-of-Lab (OOL) Glove, a low-cost motion-capture glove tailored to LaST-HD for human-hand data collection. The captured human data provide precise keypoints and serve as universal action supervision across grippers and dexterous hands. Armed with the aligned latent space and high-fidelity human-hand data, we develop a progressive mixed-to-human training recipe comprising mixed human-robot co-training and human-hand online correction post-training. Through mixed co-training, LaST-HD improves generalization to novel objects, scenes, and positions using only human-hand demonstrations. With online correction, LaST-HD further adapts to novel environments and achieves over 90% accuracy using only 20 minutes of OOL glove data.

Keywords: Robotic Manipulation, Human-Hand Data, Vision-Language-Action

1 Introduction

Developing generalist manipulation policies in unstructured environments remains a grand challenge. Driven by large-scale pretrained Vision-Language Models (VLMs) [1, 2, 3], Vision-Language-Action (VLA) models [4, 5, 6, 7] have emerged as a powerful paradigm for robotic control. However, scaling these models requires massive robot demonstration datasets [8, 9, 10], which are notoriously inefficient to collect due to the labor-intensive nature and substantial hardware overhead of real-robot teleoperation. In contrast, human hand demonstrations offer a direct, data-rich alternative with diverse physical interaction priors, providing an ideal substrate for transferring human manipulation experience to robot manipulation [11, 12].

Existing human-to-robot transfer methods have predominantly centered on kinematic retargeting [13, 14] or morphological translation [15], mapping human hand poses directly onto robotic joint spaces. Moving beyond such rigid mappings, traditional policies typically extract visual representations [16, 17] or object-centric trajectory priors [18, 12] from human videos to guide robot learning. In the VLA field, effectively leveraging large-scale human hand data has become even

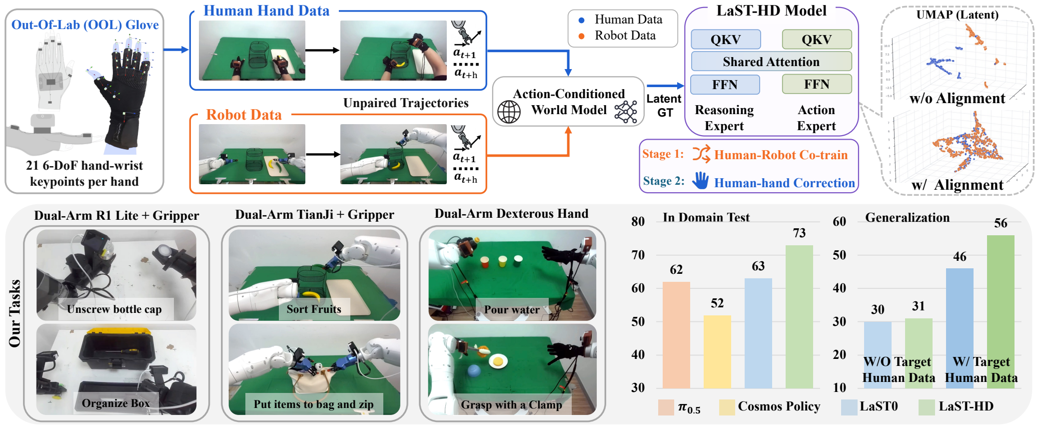

Figure 1: **Overview of LaST-HD.** LaST-HD aligns human-hand and robot demonstrations through action-conditioned latent physical reasoning, enabling morphology-agnostic action learning under a progressive mixed-to-human training recipe. To support high-quality human data collection, we introduce an OOL glove tailored to LaST-HD. We evaluate LaST-HD on six real-world tasks spanning three embodiments, including dual-arm grippers and dexterous hands.

more critical to unlocking open-world generalization [19, 20]. Recent VLA methods have begun to treat the human hand as a distinct cross-embodiment variant and rely on joint co-training to learn shared representations [21, 22, 23, 24]. Nevertheless, these co-training approaches either remain sensitive to data scale or overlook the alignment of physical dynamics between human and robot manipulation. This raises a question: Can we use the physical reasoning of VLA models as an intermediate interface to better transfer human-hand demonstrations into robot action learning?

In this paper, as shown in Figure 1, we present LaST-HD, a novel **human-to-robot action learning paradigm** that extends "reasoning-before-acting" VLA by using aligned latent reasoning as an intermediate interface for morphology-agnostic action learning. Rather than relying solely on action-level co-training, LaST-HD embeds glove-collected human hand and robot trajectories into a shared compact latent space to model their underlying physical dynamics. Building on a Mixture-of-Transformers (MoT) VLA model [25], LaST-HD performs physical reasoning in an aligned latent space and uses shared attention to guide action execution. To supervise this shared latent space, we employ an action-conditioned world model [26] trained on unpaired human and robot trajectories. Using action labels as weak anchors, the world model aligns human and robot latent features through predicted physical consequences. We then extract its predicted future features as explicit latent targets for LaST-HD's reasoning expert. This simple yet deliberate design enables our LaST-HD VLA model to reason about morphology-agnostic physical dynamics, spearheading efficient action learning from human hand data. In Figure 1, the UMAP visualizations [27] show that LaST-HD forms structured alignments between human-hand and robot trajectories from the same task.

To scale human interaction data, we develop **Out-of-Lab (OOL) Glove**, tailored to LaST-HD for low-cost and high-fidelity human-hand data collection. Unlike conventional exoskeleton gloves that mainly record device-specific joint angles, OOL Glove is designed to capture native human-hand interactions with the environment. Specifically, it uses IMU-based 6-DoF keypoint tracking to localize 21 hand-wrist keypoints with a sub-millimeter average per-keypoint RMS position error. The collected data can be retargeted from parallel grippers to dexterous hands, providing teleoperation-level supervision for robot action learning. Armed with LaST-HD's latent alignment method and high-quality human-hand data, we further propose a progressive **mixed-to-human training recipe**. Specifically, we first co-train LaST-HD on a mixed human-robot corpus, where aligned physical reasoning drives cross-embodiment action generation. As shown in Figure 1, our proposed system expands the frontier of human action learning, significantly improving generalization to novel objects, scenes, and positions using only human-hand demonstrations, without requiring target-domain robot data. After co-training, we introduce the first human-hand online correction strategy, which injects targeted human corrections at failure-prone states. By replacing cumbersome teleoperation with lightweight human-hand corrections, this post-training stage enables rapid adaptation and drives the

LaST-HD policy to over 90% accuracy in novel environments with only 20 minutes of data. **Our overall system contributions are summarized as follows:**

- We propose LaST-HD, a novel human-to-robot action learning paradigm that extends reasoning-before-acting VLA by aligning human-hand and robot data in a shared latent physical reasoning space, thereby facilitating morphology-agnostic action learning.
- We develop OOL Glove, a low-cost wearable platform tailored to LaST-HD for capturing high-quality human-hand interactions as retargetable supervision for both grippers and dexterous hands.
- We develop a mixed-to-human training recipe that activates LaST-HD's full potential by combining mixed human-robot co-training with human-hand online correction for rapid adaptation.

2 Related Work

Vision-Language-Action (VLA) Models. Driven by large-scale pretrained Vision-Language Models (VLMs) [1, 2, 3], VLA models have become a leading paradigm for generalizable robotic manipulation. Early systems such as RT-series models and OpenVLA [28, 4] demonstrated strong cross-task generalization from large-scale training data. Subsequent work improved action generation through efficient tokenization [7, 29], flow matching [5, 30, 31], and diffusion-based decoders [32, 6, 7]. More recently, "reason-before-act" VLAs have incorporated textual CoT for task planning [33, 34, 35], future visual or multimodal prediction [36, 37, 38, 39, 40, 41], and latent-space reasoning for physical dynamics modeling [42, 25, 43, 44]. Meanwhile, scaling VLA training with human hand data has emerged as a promising route toward action generalization [19, 20, 21].

Learning from Human data. Human hand demonstrations provide a scalable source of physical interaction [12, 11, 45, 46]. Early works mainly used egocentric corpora [47, 48] to learn robot visual representations [17, 49, 50]. Later methods extracted more action-proximal supervision from human videos, including keypoint or point-track prediction [18, 51], object affordances [12, 52], and latent actions [53, 54, 44]; others collected aligned human-robot action pairs with AR/VR systems [55, 56]. Recent work treats the human hand as another embodiment: EgoMimic [16] and DexWild [57] rely on explicit kinematic alignment for unified imitation learning, EgoVLA [58] transfers pretrained human-motion VLAs via inverse kinematics, while H-RDT [23] and EgoScale [24] bridge the embodiment gap through large-scale cross-embodiment pre-/mid-training. Nevertheless, these methods typically rely on action- or representation-level alignment, which can remain sensitive to data scale. In contrast, LaST-HD makes the first attempt to align human-hand and robot demonstrations through latent physical reasoning, thereby facilitating downstream action learning.

3 Methodology

In this section, we present LaST-HD framework for enhancing human hand action learning, consisting of human-to-robot alignment, the OOL Glove design, and a mixed-to-human training recipe.

3.1 Preliminaries

Problem Formulation. We formulate robotic manipulation as probabilistic sequential decision making [4]. Given a language instruction $\mathbf{l}$ and visual observation $\mathbf{I}_t \in \mathbb{R}^{H\times W\times 3}$, the policy π_θ predicts an action chunk $\mathbf{a}_{t+1:t+H} \sim \pi_\theta(\cdot \mid \mathbf{I}_t, \mathbf{l})$. The action space is embodiment-specific: dual-arm grippers use two 7-DoF end-effector actions, each consisting of relative translation, Euler-angle rotation, and a binary gripper command, while dexterous hands additionally include hand joint angles, yielding, for example, a 26-DoF action space when equipped with the 20-joint WUJI Hand.

Model Architecture. As shown in Figure 2(a), LaST-HD adopts a Mixture-of-Transformers (MoT) VLA architecture built upon Janus-Pro [59]. Following a "reasoning-before-acting" paradigm [42, 25], it first produces compact latent reasoning to model task-relevant physical dynamics before predicting robot actions. For vision encoder, each observation $\mathbf{I}_t \in \mathbb{R}^{H\times W\times 3}$ (H = W = 384), we use SigLIP-Large [60] to extract visual features $f_{\text{img}} \in \mathbb{R}^{N_{\text{img}}\times d_v}$, where N_{img} is the number of visual tokens, and d_v is the visual embedding dimension. These features are projected into the large language model (LLM) hidden space via an MLP. For the VLA backbone, we adopt DeepSeek-LLM 1.5B and repurpose its 24-layer decoder-only transformer into an MoT-based policy with two distinct components: a reasoning expert and an action expert. The reasoning expert autoregressively predicts a sequence of latent states $\mathcal{Z} \in \mathbb{R}^{N_{\text{lat}} \times d_l}$, while the action expert predicts the action chunk $\mathbf{a}_{t:t+H-1}$ through flow-matching, where N_{lat} is the number of reasoning tokens and d_l is the LLM hidden dimension. Latent reasoning knowledge is transferred from the reasoning expert to the action expert through a shared attention design. This MoT formulation provides an effective interface for injecting morphology-agnostic physical reasoning priors before action prediction.

3.2 Human-to-Robot Latent Alignment

To efficiently train LaST-HD, we leverage scalable human-hand demonstrations, yet the substantial embodiment gap between human hands and robot arms makes direct transfer challenging. Instead of naive action-level co-training, which suffers from severe domain mismatch, we introduce a human-to-robot latent alignment strategy that embeds both domains into a shared physical reasoning space, allowing aligned latent reasoning to support efficient action learning.

World Model as Alignment Bridge. To construct this shared latent space, as shown in Figure 2 (a), we fine-tune an action-conditioned world model [26], on a mixed dataset of glove-collected human and real-robot demonstrations. Importantly, demonstrations from the two domains need not be strictly paired. Specifically, we feed visual observations into the world model, injecting continuous action chunks via cross-attention into every layer to guide generation. At the final denoising step, we extract features from the deepest U-Net layer, which capture predictive physical dynamics while providing domain-invariant representations [61, 62]. These spatiotemporal features are projected to the LaST-HD latent dimension d_l with an MLP-based aligner, flattened, and compressed by adaptive average pooling into N_{lat} latent tokens. The resulting tokens serve as explicit ground-truth for supervising LaST-HD's reasoning expert. Notably, we use the world model for latent supervision rather than direct action prediction, since its latent features are not sufficiently compact for efficient control, and action conditioning may introduce information leakage.

Insights on Latent Alignment. The effectiveness of this alignment strategy stems from the physical invariance underlying manipulation tasks. Although human and robot trajectories are not explicitly paired and differ in both kinematic structure and visual appearance, action labels from both domains serve as weak anchors. Since both embodiments obey the same physical laws—for example, pushing an apple produces similar object motion regardless of embodiment—action-conditioned prediction encourages latent alignment around shared task semantics and physical dynamics. By adopting the "reasoning-before-acting" paradigm, LaST-HD projects human and robot data into a unified latent reasoning space, thereby facilitating action learning from both embodiments. To validate the effectiveness of this simple yet deliberate alignment design, we conduct UMAP visualizations and attention-map analyses on latent tokens in Appendix E and Section 4.3, respectively.

3.3 Human-Hand Data Collection

Hardware System. The Out-of-Lab (OOL) Glove is designed for low-cost, low-latency, and scalable human-hand data collection. Unlike prior exoskeleton or haptic gloves that impose rigid mechanical constraints, the OOL Glove adopts an ultra-lightweight design, with each glove weighing less than 100 g. Mechanically, as shown in Figure 2 (b), the system uses six compact IMU modules to track 20 hand keypoints and one wrist keypoint within a unified hand-centric coordinate frame. Our glove operates at over 200 Hz with less than 10 ms latency and achieves sub-millimeter average RMS position error per keypoint. Detailed specifications are provided in Appendix A.

Data Collection. During data collection, users perform manipulation tasks directly in the physical environment. As shown in Figure 2 (b), the overall collection setup includes two wrist-mounted views and one head-/chest-mounted view. More importantly, OOL Glove provides native human-hand kinematics for LaST-HD training, enabling more efficient learning of physical interactions.

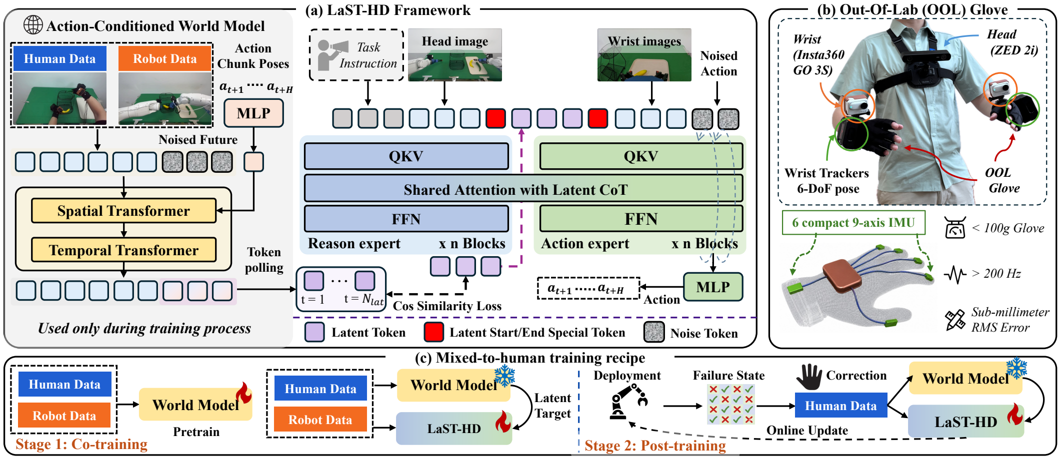

Figure 2: **Framework.** (a) LaST-HD aligns human-hand and robot data through an action-conditioned world model, which converts predicted physical consequences into denoised future-frame features as latent supervision for the reasoning expert. The aligned latent CoT conditions the action expert through shared attention for action generation. (b) We present the OOL Glove and collection platform for human-hand data collection. (c) A progressive mixed-to-human training recipe is introduced, where mixed human-robot co-training is followed by human-hand online corrections.

Compared with vision-based bare-hand tracking, this modality provides higher precision; compared with exoskeletons, OOL Glove more directly records human-hand actions, rather than indirectly recording device-specific joint angles. It also enables 4-5 $\times$ faster data collection than robot tele-operation on standard tasks, while avoiding signal loss. Each collected demonstration forms a synchronized multimodal trajectory consisting of language instructions, observations, and precise hand-wrist states. After collection, these native hand trajectories are mapped into a unified hand-centric representation. This representation serves as a general data substrate: gripper commands are derived from fingertip distances, while dexterous-hand joint angles are solved through inverse-kinematics retargeting based on the relative spatial relationships among human-hand keypoints.

3.4 Mixed-to-Human Training Recipe

As shown in Figure 2(c), we propose a progressive mixed-to-human recipe that fully activates LaST-HD's potential for learning from human-hand data.

Stage 1: Mixed Human-Robot Co-training. We first train an action-conditioned world model on mixed human-hand and robot trajectories. As described in Section 3.2, the world-model features corresponding to the predicted future frames serve as morphology-agnostic supervision for LaST-HD. We find that the world model need not be retrained for every downstream task; including target-embodiment data during pretraining is sufficient to provide aligned latent targets. For example, our pretraining mixture includes OOL Glove data and Tianji dual-arm data. During co-training, LaST-HD VLA model is optimized on both human-hand and robot trajectories. The action expert learns to generate executable robot actions using a flow matching action loss [5, 25] $\mathcal{L}_{\text{act}}$. The latent reasoning expert is supervised by the aligned latent targets extracted from the trained world model. Specifically, given the predicted latent tokens $\hat{\mathbf{z}}_t$ and the latent target $\mathbf{z}_t^{\text{GT}}$, we use a cosine similarity loss: $\mathcal{L}_{\text{latent}} = \sum_{t=1}^{N_{\text{lat}}} \left(1 - \frac{\hat{\mathbf{z}}_t \cdot \mathbf{z}_t^{\text{GT}}}{\|\hat{\mathbf{z}}_t\| \|\mathbf{z}_t^{\text{GT}}\|}\right)$. The overall LaST-HD model training objective is: $\mathcal{L}_{\text{loss}} = \mathcal{L}_{\text{act}} + \lambda \mathcal{L}_{\text{latent}}$, where λ balances action and latent supervision. We apply this objective to both pretraining and downstream fine-tuning; data usage details are provided in Appendix C.

Stage 2: Human-Hand Online Correction Post-training. After mixed co-training, we deploy LaST-HD on the real robot and perform policy rollouts to identify failure-prone states. Unlike previous post-training methods that collect additional robot teleoperation data [63], we collect human-hand corrective demonstrations at failure states using the OOL Glove. In this stage, the world model is kept frozen. To incorporate new corrective knowledge while avoiding catastrophic forgetting, we post-train LaST-HD for only 1–2 epochs with balanced replay. Each batch ($\mathcal{B}$) is constructed by

Table 1: **In-domain Results.** We compare LaST-HD against baselines across three distinct robot embodiments with six tasks. (Mix-HD) denote the model trained on a robot-human data mix.

Method	Galaxea R1 Lite		Tianji Marvin		Tianji Marvin + WUJI		Avg
	Unscrew Cap	Organize Box	Sort Fruits	Put and Zip	Pour Water	Grasp Clamp	
$\pi_{0.5}$	0.70	**0.70**	0.85	0.75	0.30	0.40	0.62
Cosmos-Policy	0.75	0.50	0.85	0.60	0.20	0.20	0.52
LaST$_0$	0.80	**0.70**	0.75	0.60	0.40	**0.50**	0.63
LaST-HD	**0.85**	**0.70**	**0.95**	**0.80**	**0.60**	0.45	**0.73**
LaST-HD (Mix-HD)	**0.85**	**0.70**	0.85	**0.80**	0.40	0.45	0.68

equally sampling from the previous data buffer $\mathcal{D}_{\text{prev}}$ and the human-hand DAgger buffer $\mathcal{D}_{\text{dagger}}$: $\mathcal{B} = \mathcal{B}_{\text{prev}} \cup \mathcal{B}_{\text{dagger}}, |\mathcal{B}_{\text{prev}}| = |\mathcal{B}_{\text{dagger}}|$. The same LaST-HD objective is used for post-training.

4 Experiments

Section 4.1 details the experimental setup. Section 4.2 presents comprehensive evaluations of LaST-HD, covering in-domain training scenarios and generalization scenarios. Finally, Section 4.3 provides ablation studies on key components of our design.

4.1 Experiment Setup

Data Collection. We evaluate LaST-HD on six real-world tasks across three embodiments. For dual-arm parallel-gripper setups, we evaluate (1) *Unscrew Bottle Cap* and (2) *Organize Box* on the Galaxea R1 Lite, alongside (3) *Sort Fruits* and (4) *Put Items to Bag and Zip* on the Tianji Marvin. For dexterous manipulation, we evaluate (5) *Pour Water* and (6) *Grasp with a Clamp* on the Marvin arm equipped with a WUJI hand. All setups utilize three 384×384 camera views: one head view (ZED 2i) and two wrist views (Insta360 GO 3S). For each task, we collect 100 in-domain robot teleoperation demonstrations and 50 OOL Glove demonstrations. We define three generalization scenarios, including unseen positions, objects, and scenes, and collect only 60 OOL Glove demonstrations per scenario for each task. More details of the hardware setup and data collection are provided in Appendix B and Appendix C, respectively.

Baselines. We compare LaST-HD with three representative SOTA methods: LaST$_0$ [25], a latent-CoT VLA model; $\pi_{0.5}$ [5], a strong VLA policy; and Cosmos-Policy [64], a world-action model. All baselines use official full fine-tuning setups; implementation details are provided in Appendix F.

4.2 Result Analysis

In-Domain Setting. We evaluate all methods in training-domain scenarios. All baselines are trained on 100 real-robot demonstrations. For LaST-HD, we evaluate two variants: LaST-HD trained with 100 robot demonstrations, and LaST-HD (Mix-HD) trained with 50 robot and 50 OOL Glove demonstrations collected under the same in-domain scenario. Each method is evaluated with 20 rollouts per task. We report task completion success rates, with success criteria detailed in Appendix C.

Table 1 reports the in-domain results, where both LaST-HD variants achieve the highest average success rates across six complex tasks. While baselines struggle on multi-step tasks such as *Sort Fruits* and *Put Items into Bag and Zip*, LaST-HD achieves 95% and 80% success rates, respectively, benefiting from stronger physical latent reasoning. The gap further widens in high-DoF dexterous manipulation with the Tianji Marvin + WUJI setup. LaST-HD (Mix-HD) maintains performance comparable to LaST-HD on four out of six tasks, indicating that OOL Glove demonstrations provide effective supervision for action learning. These results validate that LaST-HD supports complex physical reasoning across challenging tasks, while mixed-data co-training effectively injects human priors without degrading native robot capabilities.

Generalization Setting. To evaluate the generalization capacities, we assess the models under two distinct paradigms. First, in zero-shot evaluation, $\pi_{0.5}$, Cosmos-Policy, LaST$_0$, and LaST-HD (Mix-HD) are directly tested using their in-domain checkpoints without further fine-tuning.

Table 2: **Generalization Results.** Task-level success rates across three generalization scenarios, including zero-shot evaluation and adaptation using only OOL Glove human data (w/ unseen HD).

Method	Scenario	Galaxea R1 Lite		Tianji Marvin		Tianji Marvin + WUJI		Avg	Global Avg
		Unscrew Cap	Organize Box	Sort Fruits	Put and Zip	Pour Water	Grasp Clamp		
Trained on in-domain data only									
$\pi_{0.5}$	Position	0.10	0.10	0.15	0.15	0.10	0.10	0.12	0.30
	Object	0.40	0.40	0.50	**0.60**	0.15	0.10	0.36	
	Background	0.45	0.40	0.75	0.60	0.20	0.20	0.43	
Cosmos-Policy	Position	0.10	0.15	0.35	0.00	0.10	0.05	0.13	0.26
	Object	0.60	0.10	0.60	0.10	0.20	0.05	0.28	
	Background	0.60	0.15	0.65	0.60	0.15	0.10	0.38	
LaST_0	Position	0.15	0.10	0.35	0.05	0.20	0.05	0.15	0.30
	Object	0.40	0.35	0.45	0.35	0.30	0.05	0.32	
	Background	0.55	0.35	0.60	0.50	0.35	0.20	0.43	
LaST-HD (Mix-HD)	Position	0.20	0.15	0.25	0.10	0.10	0.10	0.15	0.31
	Object	0.45	0.40	0.40	0.35	0.40	0.10	0.35	
	Background	0.55	0.45	0.50	0.40	0.40	0.30	0.43	
Trained with additional human data in unseen scenarios									
LaST_0 (w/ unseen HD)	Position	**0.30**	**0.40**	0.50	0.35	0.20	0.20	0.33	0.46
	Object	0.65	0.50	0.65	0.50	0.35	0.35	0.49	
	Background	**0.65**	0.60	0.70	0.60	0.45	**0.45**	0.58	
LaST-HD (w/ unseen HD)	Position	**0.30**	**0.40**	**0.60**	**0.45**	**0.35**	**0.35**	**0.41**	**0.56**
	Object	**0.75**	**0.60**	**0.70**	**0.60**	**0.45**	**0.40**	**0.58**	
	Background	**0.65**	**0.70**	**0.90**	**0.80**	**0.60**	**0.45**	**0.68**	

Second, to assess whether human-hand data alone can enable generalization to unseen scenarios, we compare LaST_0 (w/ unseen HD) and LaST-HD (w/ unseen HD), both trained on 100 in-domain robot demonstrations plus 60 human-hand demonstrations per unseen scenario. We report only the average success rate over the three scenarios (60 rollouts); detailed results are provided in Appendix D. We visualize the real-robot evaluation process and unseen scenarios in Figure 4.

As shown in Table 2, LaST-HD (w/ unseen HD) achieves substantial improvements over all baselines, showing that aligning human and robot data in a shared latent space enables low-cost human demonstrations to drive robot generalization to new scenarios. **Unseen Position:** In the zero-shot setting, all methods show a substantial drop in average success rate, e.g., $\pi_{0.5}$ drops to 12%. With target-domain human-hand data, LaST-HD (w/ unseen HD) reaches 41% average success, demonstrating that human data helps expand the policy's physical exploration space. **Unseen Object:** LaST-HD (w/ unseen HD) improves the average success rate to 58%, outperforming the previous SOTA $\pi_{0.5}$ by 22% and showing that physical latent reasoning supports robust semantic understanding and adaptive contact patterns. **Unseen Background:** LaST-HD (w/ unseen HD) achieves a strong 68% average success across six tasks, showing that low-cost human-hand data, together with our proposed training recipe, improves robustness to novel visual contexts.

Human-Hand Online Correction. As shown in Fig. 3(a), we evaluate human-hand post-training on *Sort Fruits* incrementally: after collecting 10, 20, and 60 OOL Glove trajectories (n), we update LaST-HD and test the resulting policy at each stage. Collecting 60 OOL Glove trajectories takes only 20 minutes. With 20 human-hand trajectories, the success rate reaches 100% for Unseen Background, while with 60 human-hand trajectories, it reaches 100% for Unseen Object. Even under the hardest spatial shift, *Unseen Position*, performance improves monotonically from 60% to 80%. These results show that our method, together with OOL Glove data collection, enables rapid adaptation when robots are deployed in new scenarios, while LaST-HD's human-to-robot latent alignment improves the efficiency of learning from human demonstrations.

4.3 Ablation Study

To validate the key design choices of LaST-HD, we conduct ablations on the dual-arm Sort Fruits task. All results are averaged success rates evaluated across three generalization settings.

Effect of latent alignment strategies. We first examine how latent target design affects LaST-HD's mixed human-robot learning. As shown in the left side of Figure 3(b), we compare LaST-HD with three variants: WM-only uses world-model latent targets without action conditioning; SigLIP follows LaST_0 [25] and uses future SigLIP image features as latent targets; and W/o Latent removes latent reasoning and retains only the action expert. For results, removing latent reasoning causes a clear performance drop from 73% to 60%, showing that action-level co-training alone cannot fully

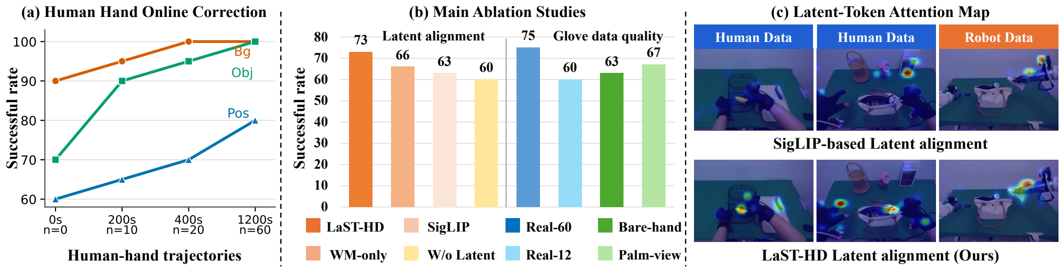

Figure 3: **(a)** Success rate under online correction with varying amounts of correction data. **(b)** Main ablation studies on LaST-HD design choices. **(c)** Visualization of latent-token attention maps.

Figure 4: Robot executions are shown on the left, and OOL Glove data in unseen scenarios on the right. Red boxes mark unseen configurations, while blue boxes mark in-domain configurations.

exploit human-hand demonstrations. Compared with the SigLIP-based and WM-only variants, using the action-conditioned world-model latents as supervision enables LaST-HD to achieve the best performance. This indicates that world-model targets capture temporal cues, while action conditioning further anchors human-hand and robot trajectories through predicted physical consequences. As shown in Fig. 3(c), LaST-HD shows stronger latent attention on embodiment-object interactions than the SigLIP-based baseline for both human-hand and robot data, indicating more task-relevant and embodiment-agnostic physical reasoning.

Effectiveness of OOL Glove data. We further evaluate the quality of OOL Glove data against other data collection paradigms under the same LaST-HD training recipe. As shown in the right part of Fig. 3(b), across three generalization scenarios, we compare 60 OOL Glove demonstrations with 60 vision-based Bare-hand demonstrations [65], as well as real-robot data using either 12 trajectories per scenario (Real-12), which matches the collection time of 60 human-hand demonstrations, or 60 trajectories per scenario (Real-60). Implementation details are provided in Appendix F. OOL Glove data achieves 73%, consistently outperforming bare-hand demonstrations (63%), indicating higher-quality interaction data. Under the same collection time, our method outperforms real-robot data (Real-12) by 15%, while remaining comparable to the Real-60 setting. This shows that our latent alignment strategy enables efficient transfer of human interaction data into robot action learning. We also find that placing the wrist camera near the thumb-index web space outperforms the palm-view setup (67%). Additional ablations on LaST-HD hyperparameters are provided in Appendix D.

5 Conclusion and Limitation

We presented LaST-HD, a reasoning-before-acting VLA framework that learns robot actions from human-hand data through human-to-robot latent alignment. By supervising latent reasoning with action-conditioned world-model features, LaST-HD captures morphology-agnostic physical dynamics and transfers human interaction knowledge to robot action generation. To support scalable and high-fidelity human data collection, we introduced OOL Glove, a low-cost wearable system for capturing native human-hand interactions. We further proposed a mixed-to-human training recipe that progressively improves LaST-HD's dynamics modeling and action generation, achieving over 90% accuracy in novel scenarios. One limitation is that latent reasoning is not yet real-time, while our main insight is to use physical reasoning to improve human action learning. Future work will explore fast-slow system designs from prior work [7] or further compress the latent space.

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A Additional Details of OOL Glove

A.1 Hardware System

A.2 Data Collection

During data collection, users perform manipulation tasks directly in the physical environment while wearing the OOL Glove. Unlike conventional robot teleoperation systems that require an intermediate human-to-robot control interface (e.g., master arms, space mice, or handheld grippers), OOL Glove directly records native human-hand motions. As a result, the collection process closely matches natural human behavior and substantially improves data collection efficiency. In practice, this allows demonstrations to be collected 4–5× faster than standard robot teleoperation while preserving rich human interaction dynamics. Meanwhile, compared with vision-based bare-hand tracking methods, OOL Glove provides more accurate and temporally stable hand-state measurements. Compared with exoskeletons or teleoperation devices, it avoids recording embodiment-specific control signals and instead directly captures human actions.

For wrist-camera placement, the camera is not restricted to a specific location and can be mounted, for example, on the underside of the wrist or the back of the hand. However, we empirically find that positioning the camera near the thumb-index web space provides the most informative viewpoint. This location offers better visibility of object contacts and finger-object interactions, particularly for dexterous manipulation tasks.

Each demonstration yields a synchronized multimodal trajectory consisting of observations, actions, and language instructions. Visual observations and hand-wrist states are recorded directly during data collection. Language instructions can be obtained either by recording spoken task descriptions through a microphone and subsequently refining the transcriptions with a vision-language model (VLM) [66], or by directly annotating demonstrations using a VLM. This process enables scalable acquisition of multimodal robot-learning data with minimal manual effort.

After collection, all demonstrations are converted into a unified hand-centric representation. This representation exhibits strong cross-embodiment reusability. For parallel-jaw grippers, grasp commands can be derived from fingertip distances and wrist trajectories. For dexterous hands, finger motions can be retargeted through kinematic mapping while preserving fine-grained contact geometry and manipulation intent. Consequently, the same human demonstration can be reused across both low-DoF grippers and high-DoF dexterous hands, significantly improving data scalability. For hand-to-robot retargeting, our current system employs heuristic mappings tailored to each dexterous hand embodiment. Consequently, a new retargeting pipeline must be constructed when introducing a new robotic hand. An important future direction is to replace these manually designed mappings with learnable retargeting models. We also observe that retargeting quality plays a critical role throughout both large-scale pre-training and downstream supervised fine-tuning, where preserving manipulation intent and contact dynamics is essential for successful policy learning.

B Real-World Setup

As shown in Figure 5, the physical deployment of LaST-HD spans three real-world robot embodiments, covering both dual-arm parallel-gripper manipulation and high-DoF dexterous control. For the 6-DoF Galaxea R1 Lite platform, we use a dual-arm configuration of right (R) and left (L) arm with parallel grippers (g) for *Unscrew Bottle Cap* and *Organize Box*. The action $\mathbf{a}_m$ is formulated as:

$$\mathbf{a}_m = [\Delta\theta^R_{1:6}, g^R, \Delta\theta^L_{1:6}, g^L] \in \mathbb{R}^{14} \tag{1}$$

For the 7-DoF Tianji Marvin parallel-gripper setup, we evaluate dual-arm manipulation on *Sort Fruits* and *Put Items to Bag and Zip*. The action $\mathbf{a}_m$ is formulated as:

$$\mathbf{a}_m = [\Delta\theta^R_{1:7}, g^R, \Delta\theta^L_{1:7}, g^L] \in \mathbb{R}^{16} \tag{2}$$

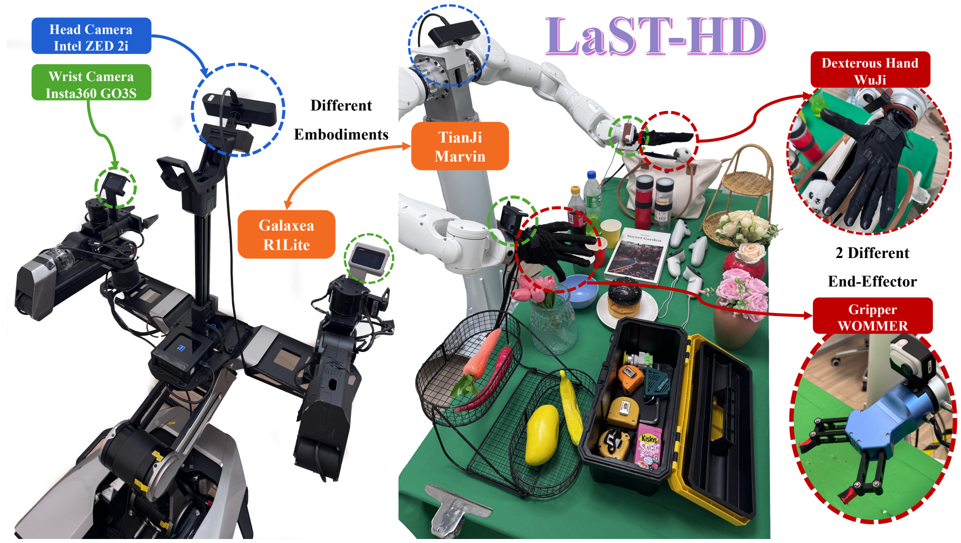

Figure 5: **Real-world setup.** Visualization of the two robot embodiments and three actuator configurations used in our real-world experiments, together with the corresponding task assets.

For dexterous manipulation, we equip a Tianji Marvin arm with a 20 DoF WUJI dexterous hand (h) and evaluate *Pour Water* and *Grasp with a Clamp*. The action $\mathbf{a}_m$ is formulated as:

$$\mathbf{a}_m = [\Delta\theta^R_{1:7}, \Delta h^R_{1:20}, \Delta\theta^L_{1:7}, \Delta h^L_{1:20}] \in \mathbb{R}^{54} \tag{3}$$

Across all platforms, the observation suite consists of three RGB views resized to 384×384: one head-mounted ZED 2i camera providing an egocentric scene view, and two wrist-mounted Insta360 GO 3S cameras capturing close-range embodiment-object interactions around the end effectors.

Robot demonstrations are collected through embodiment-specific teleoperation interfaces. For Galaxea R1 Lite, we use its native leader-follower teleoperation system, where a human-operated leader arm with matched kinematics commands the R1 Lite follower arms, enabling intuitive bi-manual data collection. On the Tianji Marvin parallel-gripper setup, operators use SpaceMouse teleoperation to provide 6-DoF Cartesian end-effector commands together with gripper open/close commands. On the Tianji Marvin + WUJI setup, operators wear our OOL Glove: wrist motion drives the Marvin arm, while finger-joint measurements are retargeted online to the WUJI hand, enabling synchronized arm-hand demonstrations. All teleoperated trajectories are synchronized with the three camera streams and stored as language-conditioned observation-action sequences for downstream training and evaluation.

C Training Datasets and Evaluation Protocols

In this section, we describe the large-scale real-robot datasets used for pre-training and the human-hand demonstrations collected with our OOL Glove. We further detail the construction of our six real-world manipulation tasks, including the task setups and success criteria used for evaluation.

C.1 Large Scale Pre-Training

To endow LaST-HD VLA model with broad motor primitives and physical common sense before downstream human-to-robot co-training [4], we adopt a large-scale pre-training mixture consisting of 400K trajectories and 28M frames curated from Open-X-Embodiment [67], DROID [8], and RoboMIND [9]. Table 3 reports the proportion of each dataset used in the pre-training mixture. Following standard data filtering practices in prior VLA works [25], we remove low-quality trajectories and reconstruct the relative action poses from consecutive robot-state differences to ensure physically consistent action annotations. Our framework is naturally compatible with both large-scale

Table 3: **Datasets for Pre-training.** The names of selected datasets for large-scale pretraining and their sampling ratios (%).

Training Dataset Mixture			
Dataset	**Ratio (%)**	**Dataset**	**Ratio (%)**
BridgeV2 [68, 69]	20.82	Nyu Franka Play [70]	0.24
Kuka [71]	20.22	Stanford Hydra [72]	0.20
Fractal [73]	13.67	RoboMIND [9]	0.20
Robo-Net [74]	11.53	Jaco Play [75]	0.19
Language Table [76]	7.72	Dobb-E [77]	0.18
BC-Z [78]	7.54	Toto [79]	0.17
Maniskill [80]	5.26	Furniture Bench [81]	0.09
DROID [8]	4.82	Utokyo Pr2 Tabletop [82]	0.04
Roboset [83]	3.21	Utokyo Xarm Pap [84]	0.04
FMB Dataset [85]	1.50	CMU Stretch [86]	0.02
Taco Play [87, 88]	1.26	DLR Sara Grid Clamp [89]	0.02
RoboTurk [90]	0.70	Utokyo Pr2 Fridge [82]	0.01
Berkeley Autolab Ur5 [91]	0.35	—	—

robot demonstrations and human-hand demonstrations collected by OOL Glove. In principle, both the action-conditioned world model and the LaST-HD VLA model can be pre-trained on the above robot datasets together with our collected 500 hours of OOL Glove human-hand data. However, to ensure fair comparison with prior VLA baselines [5, 25], we pre-train the LaST-HD Mixture-of-Transformers (MoT) model only on real-robot trajectories.

To provide explicit supervision for latent physical reasoning, we first pre-train an action-conditioned world model using both OOL Glove human-hand demonstrations and real-robot trajectories. The world model takes visual observations as input and injects the corresponding action chunks through cross-attention, learning to predict future physical consequences across human and robot embodiments. After pre-training, we freeze the world model and use it to precompute latent ground-truth targets for LaST-HD. Specifically, for each trajectory, we extract features from the deepest U-Net layer at the final denoising step, where the features encode action-aware spatiotemporal dynamics. These features are then projected into the LaST-HD latent dimension with an MLP aligner, flattened, and compressed through adaptive average pooling into a fixed number of latent tokens. The impact of the latent-token length is further analyzed through an ablation study in Appendix D. The resulting tokens serve as latent ground-truth targets for supervising the reasoning expert of LaST-HD.

Because these latent targets are computed entirely offline, they are simply loaded together with the standard visual, language, and action inputs during training. This design provides LaST-HD with compact action-conditioned physical reasoning anchors, enabling the model to learn predictive environmental dynamics and establish a shared latent reasoning space across human and robot data. We are continuously collecting in-the-wild OOL Glove demonstrations and plan to release a high-quality human-hand manipulation dataset to support future research on scalable human-to-robot action learning.

C.2 Downstream Task Descriptions and Success Criteria

Task 1: Unscrew Bottle Cap. The robot first stabilizes a bottle using its left gripper while the right gripper performs repeated rotational motions to unscrew the bottle cap. After the cap is loosened, the right gripper places it onto the tabletop. A trial is considered successful if the robot completes the cap-unscrewing motion with the right gripper.

Task 2: Organize Box. The robot organizes multiple tabletop objects into designated compartments of a toolbox using both grippers. After placing all objects, the robot closes the intermediate toolbox cover by grasping and repositioning it to the correct location. A trial is considered successful if all objects are placed into their designated compartments and the intermediate toolbox cover is correctly grasped.

Task 3: Sort Fruits. The left gripper stabilizes a two-layer storage basket while the right gripper sorts fruits and vegetables into different layers of the basket. Fruits are placed in one layer and vegetables in the other. A trial is considered successful only if all objects are placed inside the basket and each object is sorted into its designated layer.

Task 4: Put Items to Bag and Zip. The robot first picks up a tabletop object and places it into a backpack. It then grasps the right side of the backpack with one gripper while the other gripper grasps the zipper and closes the bag. A trial is considered successful if the object is successfully placed inside the backpack and the zipper is fully closed.

Task 5: Pour Water. A dexterous hand grasps a bottle and pours its contents into a target cup specified by the instruction. To ensure safe and repeatable evaluation, the bottle cap remains closed during testing, and the task focuses on the pouring motion rather than liquid transfer. A trial is considered successful if the dexterous hand successfully grasps the bottle and executes the complete pouring motion.

Task 6: Grasp with a Clamp. The dexterous hand manipulates a food clamp to grasp a meat patty, lifts it from the workspace, and transfers it onto a bread bun. A trial is considered successful if the meat patty is successfully transferred and placed on top of the bread bun.

D Additional Quantitative Results

D.1 Additional Ablation Studies

Unless otherwise specified, all ablations are conducted on the dual-arm *Sort Fruits* task. We choose this task because it requires non-trivial bimanual coordination and multi-step object manipulation, making it highly representative of LaST-HD's reasoning and control capabilities. At the same time, compared with dexterous-hand tasks, it avoids introducing additional confounding factors arising from high-dimensional hand control, providing a cleaner evaluation of the proposed design choices.

Effect of world-model denoising steps. We first investigate how the denoising step used for latent target extraction affects the final policy performance. Specifically, when constructing latent ground-truth targets with the action-conditioned world model, we extract latent features after 2, 5, and 10 denoising steps, respectively. As shown in Figure X (a), all three settings achieve similar downstream success rates, indicating that the latent supervision remains largely stable across different denoising depths. Given the negligible performance difference, we adopt 2 denoising steps throughout all experiments. Although the world model is only used offline for latent target generation and is not involved during LaST-HD VLA model inference, reducing the denoising steps significantly improves the efficiency of training dataset construction.

Effect of LaST-HD's shared latent length. We further study the impact of the shared latent length in LaST-HD. Specifically, we evaluate latent lengths of 4, 8, 12, and 16 tokens while keeping all other training configurations fixed. As shown in Figure X (b), latent lengths of 4 and 8 achieve nearly identical performance. When the latent length is increased to 12 or 16, we observe a modest improvement in task success rate, suggesting that additional latent tokens help capture longer-horizon physical dynamics. However, because latent generation inherits the native autoregressive decoding paradigm of LLMs, increasing the latent sequence length directly increases inference latency. Considering the trade-off between performance and efficiency, we adopt a latent length of 4 in all experiments.

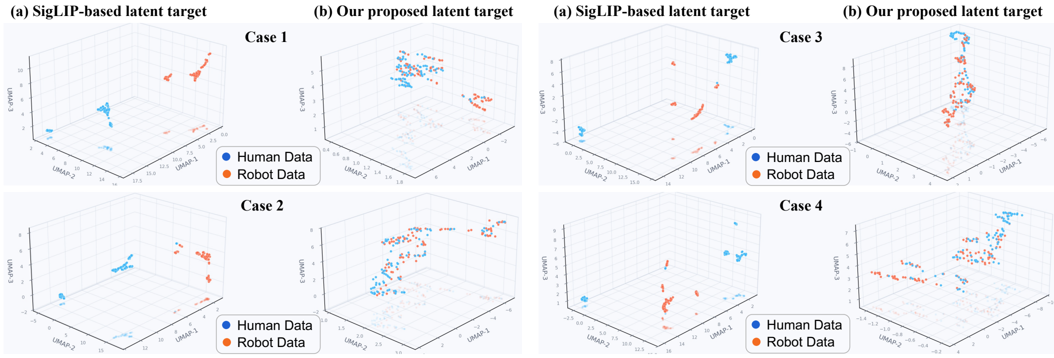

Figure 6: UMAP visualization of LaST-HD latent tokens generated by different model variants. We compare latent representations learned from (a) SigLIP-based latent targets and (b) our action-conditioned world-model latent targets.

E Additional Qualitative Results

E.1 Additional UMAP Visualization

To better understand how LaST-HD aligns human-hand demonstrations with robot trajectories, we provide additional UMAP visualizations of the learned latent representations in Figure X. Following prior work [92], we project the latent tokens produced by the reasoning expert into a three-dimensional space using UMAP [27]. Specifically, we separately collect latent representations from human-hand inputs and robot inputs under the same task and visualize them jointly.

As shown in Figure 6, LaST-HD consistently produces structured alignment between human-hand and robot trajectories belonging to the same task. Rather than forming two isolated clusters corresponding to human hand and robot, the latent representations exhibit overlapping geometric structures. This observation suggests that the proposed shared latent reasoning space successfully bridges the embodiment gap between human demonstrations and robot executions. Compared with SigLIP-based latent targets, the action-conditioned world-model supervision encourages latent representations to capture shared physical dynamics and task-relevant interaction patterns. As a result, human-hand demonstrations and robot trajectories become more closely aligned in the latent space, enabling more effective transfer of manipulation knowledge during mixed human-robot training. This structured alignment directly contributes to the strong data efficiency of LaST-HD when leveraging human-hand demonstrations.

E.2 Additional Attention Map Visualization.

To explicitly demonstrate how the proposed latent alignment facilitates action generation, we visualize the attention maps between LaST-HD output tokens and visual tokens in Figure 7. The resulting attention distributions are projected back to the image space for visualization.

The attention maps further support the UMAP analysis by showing that LaST-HD focuses primarily on embodiment-object interactions and manipulation-relevant physical dynamics. Across both human-hand and robot observations, the latent tokens consistently attend to manipulated objects, contact regions, and interaction points, while suppressing irrelevant background distractions. In contrast, SigLIP-based latent supervision tends to allocate attention more broadly across the scene and is less concentrated on the manipulated object. These visual findings elucidate the mechanism behind our shared latent reasoning: by constraining latent representations to capture task-relevant dynamics rather than domain-specific visual appearances, LaST-HD establishes a domain-invariant reasoning space that effectively harnesses human demonstrations to enhance robot policy learning.

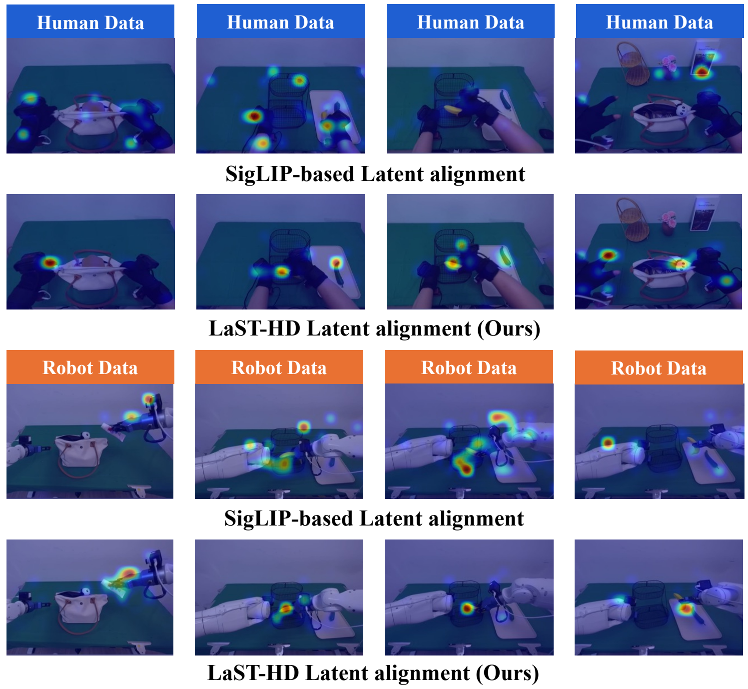

Figure 7: Visualization of LaST-HD output-token-to-visual-token attention maps.

E.3 Additional Visualizations of Robot Executions and OOL Glove Data

F Baseline Implementation Details

F.1 Model Baselines

$\pi_{0.5}$ **[33]** is a PaliGemma-based [3] vision-language-action (VLA) model that employs flow matching for action generation. We adopt full-parameter fine-tuning of the entire model during downstream post-training. Following its official open-source implementation, all input images are resized to 224×224. We keep this resolution unchanged, since modifying the image input size may disrupt the visual representations learned during pre-training.

Cosmos-Policy [64] adapts a pretrained Cosmos-Predict video foundation model into a robot policy through post-training on robot demonstrations. It represents actions, future states, and value estimates as latent frames within the video diffusion process, enabling visuomotor control. Following the official input image resolution, we resize all input images to 224×224.

LaST$_0$ **[25]** is a Janus-Pro-based [59] VLA model that introduces a Mixture-of-Transformers architecture for latent spatio-temporal Chain-of-Thought reasoning over future visual, 3D, and proprioceptive states. Following the official setup, all input images are resized to 384×384.

For all model baselines, when training on high-DoF dexterous-hand tasks, we directly modify the action output dimension to match the corresponding dexterous action space while keeping the remaining training configurations unchanged.

F.2 Data Collection Baselines

UMI [93] collects data using portable hand-held grippers equipped with cameras, allowing humans to directly demonstrate manipulation skills in the wild. Following official setups, we use three GoPro cameras to capture multi-view visual observations during data collection. The demonstrations include visual observations and 6-DoF gripper trajectories, which are then converted into robot-executable action data for policy learning.

Hawor [65] collects human-hand demonstrations from egocentric videos and reconstructs world-space hand trajectories through hand-pose estimation and camera-motion tracking. By combining adaptive egocentric SLAM with motion reconstruction, HawOR produces temporally consistent hand trajectories that can be retargeted to robot embodiments for imitation learning.

G Failure Case Analysis

Although LaST-HD achieves strong performance across different embodiments and tasks, execution failures still occur in a small number of complex scenarios, as shown in Figure X. These failures typically arise from hardware execution errors or spatial coordination challenges. The main failure cases are summarized below:

1) Object slipping during transport: Although the model correctly plans the approach and lifting trajectory, the selected grasping location can occasionally be slightly suboptimal. As a result, the object is not firmly held and may slip out of the gripper, causing task failure. Although our model has a certain degree of closed-loop recovery capability and may attempt to grasp the object again, this substantially reduces task efficiency. Moreover, if the object falls into an unrecoverable position or state, the task ultimately fails.

2) Collision between dual-arm wrist cameras: For example, in the dual-arm *Put Items to Bag and Zip* task, both arms must operate in a highly constrained and spatially close region. When one arm attempts to grasp the zipper, its target position can be very close to the other arm. Since the camera viewpoints in human-hand data are not precisely aligned with those of the robot wrist cameras, introducing human-hand data can occasionally lead to collisions between the two wrist cameras during robot inference. In future work, we plan to incorporate automatic collision-avoidance strategies.

3) Object collision under cluttered backgrounds: In highly cluttered unseen scenes, due to environmental changes, the manipulated object may occasionally collide with nearby objects in the new environment while being moved by the robot arm. In future work, we will collect more OOL Glove demonstrations in diverse environments for pre-training, thereby improving the robustness of motion planning.